

Fujitsu FM-NEW7 ICs

Part ID	Part	Equivalent	Function	Usage
M1	M74LS273P	74LS273	Octal D flip-flop with clear	
M2	MB74LS05	74LS05	Hex inverter, open-collector outputs	
M3	MB74LS245	74LS245	Octal bus transceiver, 3-state	
M4	MB74LS244	74LS244	Octal buffer / line driver, 3-state, non-inverting	
M5	MB74LS14	74LS14	Hex Schmitt-trigger inverter	
M6	MB74LS04	74LS04	Hex inverter	
M7	MB74LS244	74LS244	Octal buffer / line driver, 3-state, non-inverting	
M8	MB74LS14	74LS14	Hex Schmitt-trigger inverter	
M9	MB74LS14	74LS14	Hex Schmitt-trigger inverter	
M10	MB74LS367A	74LS367A	Hex bus driver, 3-state (4+2 enable)	
M11	SN7407N	7407	Hex buffer / driver, open-collector, high-voltage outputs	
M12	MB74LS74A	74LS74A	Dual D flip-flop with preset and clear	
M13	MB74LS02	74LS02	Quad 2-input NOR gate	
M14	386D	LM386 / NJM386D	Low-voltage audio power amplifier	Internal speaker output stage - amplifies beep and PSG audio
M15	SN7407N	7407	Hex buffer / driver, open-collector, high-voltage outputs	
M16	SN7407N	7407	Hex buffer / driver, open-collector, high-voltage outputs	
M17	MB3614	LM324	Quad operational amplifier, 14-pin	Analogue signal conditioning - cassette (CMT) read amplifier / comparator and audio mixing
M18	MB74LS08	74LS08	Quad 2-input AND gate	
M19	MB74LS175	74LS175	Quad D flip-flop with clear	
M20	MB74LS00	74LS00	Quad 2-input NAND gate	
M21	MB74LS244	74LS244	Octal buffer / line driver, 3-state, non-inverting	
M22	MB74LS74A	74LS74A	Dual D flip-flop with preset and clear	
M23	MB74LS175	74LS175	Quad D flip-flop with clear	
M24	MB74LS367A	74LS367A	Hex bus driver, 3-state (4+2 enable)	
M25	MB74LS244	74LS244	Octal buffer / line driver, 3-state, non-inverting	
M26	MB74LS244	74LS244	Octal buffer / line driver, 3-state, non-inverting	
M27	MB74LS374	74LS374	Octal D flip-flop, 3-state outputs	
M28	AY-3-8910	AY-3-8910	3-voice PSG sound generator with two 8-bit parallel I/O ports	Built-in sound, on the sub-CPU side; PSG I/O ports serve the joystick interface
M29	MB15021		Custom digital colour palette LSI	Maps the 8 colour codes (0-7) to RGB output
M30	MB74LS132	74LS132	Quad 2-input NAND Schmitt trigger	
M31	MB84049B	CD4049B	CMOS hex inverting buffer / level shifter	Unknown - conventionally level translation on the cassette or external interface
M32	MB434	SN75452 (?)	Dual peripheral / power driver, 8-pin, open-collector outputs	Ch.1 (pin 3) drives the buzzer; Ch.2 (pin 5) drives the cassette motor relay (??)
M33	MB8265A-15	4164	64K x 1 bit DRAM, 150ns	Main RAM (8 devices = 64KB)
M34	MB74LS240	74LS240	Octal buffer / line driver, 3-state, inverting	
M35	MB74LS138	74LS138	3-to-8 line decoder / demultiplexer	
M36	HD68B09P	MC68B09	8-bit CPU, 2MHz rated	Sub-CPU - display, VRAM, keyboard and PSG
M37	MB74LS245	74LS245	Octal bus transceiver, 3-state	
M38	MB112T605		Fujitsu semi-custom gate array	Unknown
M39	MB60H010A		Custom 64-pin LSI	Likely the display controller (the FM-7 has no standard 6845)
M40	MB74LS244	74LS244	Octal buffer / line driver, 3-state, non-inverting	
M41	MB8128-15	6116	2K x 8 bit SRAM, 150ns	Sub-system work RAM / inter-CPU shared RAM
M42	MB8265A-15	4164	64K x 1 bit DRAM, 150ns	Main RAM (8 devices = 64KB)
M43	MB8265A-15	4164	64K x 1 bit DRAM, 150ns	Main RAM (8 devices = 64KB)
M44	MB74LS04	74LS04	Hex inverter	
M45	MB74LS04	74LS04	Hex inverter	
M46	MB74LS138	74LS138	3-to-8 line decoder / demultiplexer	
M47	MB74LS244	74LS244	Octal buffer / line driver, 3-state, non-inverting	
M48	MB74LS244	74LS244	Octal buffer / line driver, 3-state, non-inverting	
M49	MB112T604		Fujitsu semi-custom gate array	Unknown
M50	MB81416-12	4416	16K x 4 bit DRAM, 120ns	Graphics VRAM (6 devices = 48KB)
M51	MB81416-12	4416	16K x 4 bit DRAM, 120ns	Graphics VRAM (6 devices = 48KB)
M52	MB8128-15	6116	2K x 8 bit SRAM, 150ns	Sub-system work RAM / inter-CPU shared RAM
M53	MB8265A-15	4164	64K x 1 bit DRAM, 150ns	Main RAM (8 devices = 64KB)
M54	HD68B09P	MC68B09	8-bit CPU, 2MHz rated	Main CPU - runs the boot ROM and F-BASIC
M55	MB112T603		Fujitsu semi-custom gate array	Unknown
M56	MB74LS05	74LS05	Hex inverter, open-collector outputs	
M57	MB74LS244	74LS244	Octal buffer / line driver, 3-state, non-inverting	
M58	MB81416-12	4416	16K x 4 bit DRAM, 120ns	Graphics VRAM (6 devices = 48KB)
M59	MB81416-12	4416	16K x 4 bit DRAM, 120ns	Graphics VRAM (6 devices = 48KB)
M60	MB83256 (167)	32K x 8 mask ROM (27256 footprint)	32KByte mask ROM	Character generator & sub-system monitor
M61	MB8265A-15	4164	64K x 1 bit DRAM, 150ns	Main RAM (8 devices = 64KB)
M62	MB14415		Unknown	Unknown
M63	MB74LS138	74LS138	3-to-8 line decoder / demultiplexer	
M64	MB74LS158	74LS158	Quad 2-to-1 multiplexer, inverting outputs	
M65	MB74LS158	74LS158	Quad 2-to-1 multiplexer, inverting outputs	
M66	MB74LS245	74LS245	Octal bus transceiver, 3-state	

Part ID	Part	Equivalent	Function	Usage
M67	MB74LS244	74LS244	Octal buffer / line driver, 3-state, non-inverting	
M68	MB81416-12	4416	16K x 4 bit DRAM, 120ns	Graphics VRAM (6 devices = 48KB)
M69	MB81416-12	4416	16K x 4 bit DRAM, 120ns	Graphics VRAM (6 devices = 48KB)
M70	MB74LS08	74LS08	Quad 2-input AND gate	
M71	MB8265A-15	4164	64K x 1 bit DRAM, 150ns	Main RAM (8 devices = 64KB)
M72	MB111S601		Fujitsu semi-custom gate array	Unknown
M73	MB8516	2716	2K x 8 EPROM / mask ROM	Boot ROM (four 512 byte blocks)
M74	MB83256 (160)	32K x 8 mask ROM (27256 footprint)	32KByte mask ROM	F-BASIC V3.02
M75	MB74LS30	74LS30	8-input NAND gate	
M76	MB88401	MB8840-series 4-bit MCU	Single-chip 4-bit microcontroller with internal mask ROM	Keyboard controller - key matrix scanning, key-code generation and LED control
M77	MB8265A-15	4164	64K x 1 bit DRAM, 150ns	Main RAM (8 devices = 64KB)
M78	MB8265A-15	4164	64K x 1 bit DRAM, 150ns	Main RAM (8 devices = 64KB)
M79	MB112T602		Fujitsu semi-custom gate array	Unknown
M80	MB112T601		Fujitsu semi-custom gate array	Unknown
M81	MB74LS245	74LS245	Octal bus transceiver, 3-state	
M82	MB74LS244	74LS244	Octal buffer / line driver, 3-state, non-inverting	
M83	MB8128-15	6116	2K x 8 bit SRAM, 150ns	Sub-system work RAM / inter-CPU shared RAM
M84	MB74LS145	74LS145	BCD-to-decimal decoder / driver, open-collector, high-voltage	
M85	MB74LS145	74LS145	BCD-to-decimal decoder / driver, open-collector, high-voltage	

Table 1

Part ID	Part			
C1	100pF (101) ceramic capacitor			
C2	100pF (101) ceramic capacitor			
C3	100pF (101) ceramic capacitor			
C4	100pF (101) ceramic capacitor			
C5	100pF (101) ceramic capacitor			
C6	100pF (101) ceramic capacitor			
C7	22nF (222) ceramic capacitor			
C8	47nF (473) ceramic capacitor			
C9	100µF 25V axial electrolytic			
C10				
C11				
C12				
C13				
C14				
C15	100µF 25V axial electrolytic			
C16				
C17				
C18				
C19				
C20				
C21				
C22				
C23	100µF 25V axial electrolytic			
C24	47µF 25V axial electrolytic			
C25	47µF 25V axial electrolytic			
C26	100µF 25V axial electrolytic			